



APPENDIX A

M68000 FAMILY SUMMARY

This Appendix summarizes the characteristics of the microprocessors in the M68000 Family. Refer to M68000 PM AD, *M68000 Programmer's Reference Manual*, for more detailed information about MC68000 and MC68010 differences.

	MC68000	MC68008	MC68010	MC68020	MC68030
Data Bus Size (Bits)	16	8	16	8,16,32	8,16,32
Address Bus Size (Bits)	24	20	24	32	32
Instruction Cache (in words)	—	—	3 ¹	128	128
Data Cache (in words)	—	—	—	—	128

Note 1. The MC68010 supports a three-word cache for the loop mode.

Virtual Memory/Machine

MC68010, MC68020, and MC68030	Provide Bus Error Detection, Fault Recovery
MC68030	On-Chip MMU

Coprocessor Interface

MC68000, MC68008, and MC68010	Emulated in Software
MC68020 and MC68030	In Microcode



Word/Long-Word Data Alignment

MC68000, MC68008, and MC68010	Word/Long Data, Instructions, and Stack Must be Word Aligned
MC68020 and MC68030	Only Instructions Must be Word Aligned (Data Alignment Improves Performance)

Control Registers

MC68000 and MC68008	None
MC68010	SFC, DFC, VBR
MC68020	SFC, DFC, VBR, CACR, CAAR
MC68030	SFC, DFC, VBR, CACR, CAAR, CRP, SRP, TC, TT0, TT1, PSR

Stack Pointers

MC68000, MC68008, and MC68010	USP, SSP
MC68020 and MC68030	USP, SSP (MSP, ISP)

Status Register Bits

MC68000, MC68008, and MC68010	T, S, I0/I1/I2, X/N/Z/V/C
MC68020 and MC68030	T0/T1, S, M, I0/I1/I2, X/N/Z/V/C



Function Code/Address Space

MC68000 and MC68008	FC0-FC2 = 7 is Interrupt Acknowledge Only
MC68010, MC68020, and MC68030	FC0-FC2 = 7 is CPU Space

Indivisible Bus Cycles

MC68000, MC68008, and MC68010	Use $\overline{AS}$ Signal
MC68020 and MC68030	Use $\overline{RMC}$ Signal

Stack Frames

MC68000 and MC68008	Support Original Set
MC68010	Supports Formats \$0, \$8
MC68020 and MC68030	Support Formats \$0, \$1, \$2, \$9, \$A, \$B

Addressing Modes

MC68020 and MC68030 extensions:	Memory indirect addressing modes, scaled index, and larger displacements. Refer to specific data sheets for details.
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MC68020 and MC68030 Instruction Set Extensions

Bcc	Supports 32-Bit Displacements
BFxxxx	Bit Field Instructions (BFCHG, BFCLR, BFEXTS, BFEXTU, BFFFO, BFINS, BFSET, BFTST)
BKPT	New Instruction Functionality
BRA	Supports 32-Bit Displacements
BSR	Supports 32-Bit Displacements
CALLM	New Instruction (MC68020 only)
CAS, CAS2	New Instructions
CHK	Supports 32-Bit Operands
CHK2	New Instruction
CMPI	Supports Program Counter Relative Addressing Modes
CMP2	New Instruction
cp	Coprocessor Instructions
DIVS/DIVU	Supports 32-Bit and 64-Bit Operands
EXTB	Supports 8-Bit Extend to 32 Bits
LINK	Supports 32-Bit Displacement
MOVEC	Supports New Control Registers
MULS/MULU	Supports 32-Bit Operands
PACK	New Instruction
PFLUSH	MMU Instruction (MC68030 only)
PLOAD	MMU Instruction (MC68030 only)
PMOVE	MMU Instruction (MC68030 only)
PTEST	MMU Instruction (MC68030 only)
RTM	New Instruction (MC68020 only)
TST	Supports Program Counter Relative Addressing Modes
TRAPcc	New Instruction
UNPK	New Instruction

